# LSV Sample-Size Study — Results

## 1. Market model: Heston + leverage

- Base: Heston SV, variance leg unchanged (exact CIR)
- Variance leg:

$$dV_t = \kappa(\theta - V_t)dt + \xi\sqrt{V_t}dW_t^V$$

- Stock leg:

$$dS_t = L(t, S_t)\sqrt{V_t}S_t\left(\rho dW_t^V + \sqrt{1-\rho^2}dW_t^\perp\right)$$

- Leverage function (CEV skew × time-growth):

$$L(t, S) = (S/S_0)^\beta(1 + \kappa_L t)$$

- Leverage capped:

$$L \leftarrow \text{clip}(L, 0, \text{leverage_cap})$$

  — prevents blowup as $S_t \to 0$

### 1.1. Parameters

Risk-neutral-recalibrated Heston core ($\kappa = 3.0$, vol_of_variance= 0.4, $\rho = -0.7$, $\theta = v_0 = 0.04$), 3-tier leverage-shift ladder (weak/medium/hard = 1.5x/2x/3x the in-dist $(\beta, \kappa_L)$ offset from zero).

| param | in-dist | weak | medium | hard |
|---|---|---|---|---|
| $\beta$ | $-0.3$ | $-0.45$ | $-0.6$ | $-0.9$ |
| $\kappa_L$ | 0.5 | 0.75 | 1.0 | 1.5 |
| leverage_cap | 5.0 | 5.0 | 5.0 | 5.0 |
| n_substeps (fine steps/hedge) | 10 | 10 | 10 | 10 |
| n_time (hedges) | 100 | 100 | 100 | 100 |
| horizon | 100/365 | 100/365 | 100/365 | 100/365 |

## 2. Objectives

| method | hyperparams swept | extra hyperparam | fixed |
|---|---|---|---|
| `CVaR` | `learning_rate`, `alpha` | — | plain nominal CVaR-95 |
| `CVaR+augmentation` | `learning_rate`, `alpha` | `proposal_scale` | `proposal_samples=32` |
| `WDRO` | `learning_rate`, `alpha` | `rho` | `kl_epsilon=0.03`, `ula_steps=0`, `cost_power=2`, `proposal_samples=32` |
| `WDRO+convex order` | `learning_rate`, `alpha` | `rho` | as `WDRO`, plus trainable-$\lambda_{ij}$ convex-order (no-calendar-arbitrage) penalty on proposal paths |

- `batch_size=64` fixed across all 4 methods, all $n$ (not swept)
- `epsilon=0.0` (no transaction cost), `latent_dim=0` (plain per-timestep MLP+BN policy) for this run

## 3. Study design

- $n \in \{100, 250, 500, 750, 1000\}$ training paths, 3 seeds
- $n$ samples are split in train/validation (80/20) for hyperparam tuning, $k = 3$ folds
- Winning hyperparam's model reused directly (no separate full-$n$ refit)
- Test: $n_{\text{test}} = 20000$ paths, in-dist and 3-tier shifted leverage params
- Canonical metric: empirical CVaR-95 of realized pathwise loss

## 4. Results (mean over 3 seeds)

Columns: val = k-fold validation CVaR-95, test = in-dist test CVaR-95, shift = hard-tier (3x offset) shifted test CVaR-95, the worst of the 3 shift tiers. Full weak/medium/hard breakdown in the note below.

| method | n = 250 | | | n = 500 | | | n = 750 | | | n = 1000 | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | val | test | shift | val | test | shift | val | test | shift | val | test | shift |
| `CVaR` | 18.20 | 44.51 | 60.22 | 22.78 | 29.71 | 37.43 | 18.04 | 23.39 | 29.49 | 19.83 | 24.11 | 29.46 |
| `CVaR+augmentation` | 20.24 | 41.99 | 65.09 | 27.03 | 31.88 | 41.62 | 23.73 | 29.46 | 36.56 | 23.04 | 323.07 | 327.56 |
| `WDRO` | 18.54 | 37.76 | 41.46 | 23.10 | 25.19 | 32.08 | 19.48 | 22.82 | 26.90 | 21.86 | 23.54 | 27.24 |
| `WDRO+convex order` | 20.99 | 30.89 | 35.63 | 22.62 | 33.65 | 37.63 | 22.13 | 25.19 | 28.79 | 21.19 | 23.82 | 28.14 |

- `CVaR+augmentation` at $n = 1000$: one of the 3 seeds diverged on the shift tiers (test CVaR-95 in the hundreds vs. the other two seeds' $\sim 30$); mean above is skewed by that outlier, not representative of typical-seed behavior at that $n$.
- `WDRO+convex order` is competitive with plain `WDRO` in-dist and on the hard shift tier at $n \geq 750$, but noisier and slightly worse at $n = 500$; the convex-order penalty does not yet show a clear win over unconstrained `WDRO` at this sample-size range.